

Can a Fisher/Cramér-Rao Bound Predict θ 's Empirical Recovery Scatter?

Full-covariance CRB vs. real pixel-likelihood fits, aberrated wavefront

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1 Motivation

Looking at 2D residual plots of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$ for the `confocal_random_zernike` (aberrated-wavefront) dataset, the scatter looks Gaussian. That raises the obvious question: can a Fisher-information / Cramér-Rao bound predict that spread analytically, rather than only measuring it empirically?

This was tried once before and got stuck. From `notes/joint_profile_inference.tex` (a sibling investigation, §2026-07-20): a Laplace approximation using the *diagonal* of a local finite-difference Hessian at θ 's MAP was found **66–220× tighter** than the true empirical scatter across 50 shots. That work diagnosed why, and flagged two things that were never actually done:

1. θ 's failure mode is a *ridge* – a continuous, weakly-identified direction (near-zero Fisher eigenvalue), not discrete multimodality like ϕ 's. A ridge is compatible with local Fisher/CRB validity *if* it is close to linear/quadratic near the truth – but whether it actually is was flagged as unchecked (Future Work item 0a in that file) and never verified.
2. The failed attempt used a **diagonal-only** Hessian, discarding exactly the off-diagonal correlations a ridge produces. `crb_signal.py` already has `build_joint_fisher()` (full 17×17 joint Fisher over $(\phi, \theta_{z0}, \theta_{z100})$, pixel-likelihood mode, all cross terms) and `full_per_shot_covariance()` (inverts the *full* matrix) – but the latter was dead code, never called from that module's own driver. Nobody had tried the full-covariance version.

This note finishes that specific, previously-flagged, previously-skipped check, reusing the existing (but previously unwired) machinery rather than re-deriving anything. Scope, by explicit direction: **one dataset** (the aberrated `confocal_random_zernike` case), matching CRB to empirical data. A Gaussian-beam-vs-aberrated comparison, and the downstream connection to the β (signal) CRB, are deliberately deferred.

2 Setup

Dataset: `data/R20_N200_A1000000...psmapconfocal_random_zernike` (20 runs $\times$ 200 shots, 10^6 atoms/shot, aberrated wavefront via `phase_space_grids.py`'s interpolated-PSMAP path). Cloud prior: independent Gaussian on each of the 8 kinematic components (position, velocity, position spread, velocity spread, in x and y), $\sigma = 10 \mu\text{m}$ (or the matching velocity unit) for all 8. Evaluator geometry: **exactly** what `non_phase_shear/analysis/generate_kinematic_estimates.py`'s **best** (pixel-likelihood) fit uses – `bins=32`, tight half-range $1.54 \times 10^{-3} \text{ m}$, `pixel_n_gh=8` – so the CRB is evaluated on the identical pixel/bin configuration that produced the empirical numbers it is compared against. Using a different geometry would make the comparison apples-to-oranges before it starts.

All CRB quantities below are evaluated at the *true* $(\phi, \theta_{z0}, \theta_{z100})$ for each shot – the standard Cramér-Rao convention, and also convenient here since it needs no extra optimizer run (the empirical $\hat{\theta}_{\text{best}}$ values already come from a separate, already-run fit).

Implementation: `python-scripts/theta_crb_vs_empirical.py` (new), importing `crb_signal.py`'s Fisher-matrix code rather than duplicating it.

3 Step 1: Is the ridge direction actually linear/quadratic?

For 10 real shots (`run_000`), built the full 17×17 joint Fisher matrix H at the true $(\phi, \theta_{z0}, \theta_{z100})$, found its smallest eigenvalue $\lambda_{\min}$ and eigenvector v (the ridge direction; $\sigma_{\text{ridge}} = 1/\sqrt{\lambda_{\min}}$), and

profiled the *real* Poisson NLL along v at $k\sigma_{\text{ridge}}$ for $k \in \{0.5, 1, 2, 4, 8\}$, comparing to the Fisher-implied quadratic $\frac{1}{2}\lambda_{\min}(k\sigma_{\text{ridge}})^2$.

Result: not cleanly quadratic. The ratio (actual ΔNLL / quadratic ΔNLL) at $k = 0.5$ – 1 (the scale that actually matters – real fluctuations live near 1σ , not 8σ) ranged from -2.9 to $+4.3$ across the 10 shots – including several shots where the sign itself flips (the true NLL is briefly *lower*, not higher, than at the anchor point, in the direction the Fisher matrix calls “increasing”). The ratio does converge toward 1 at larger k (e.g. 0.80–1.23 at $k = 8$), consistent with the quadratic approximation eventually dominating far from the anchor, but that is not the regime real shot noise samples.

This is a genuine, unresolved finding, not swept under the rug: the single worst-conditioned direction is *not* well-approximated by the local quadratic at the scale that matters. And yet – see §4 – the *aggregate*, full 8-dimensional marginal covariance built from the same local Fisher matrix matches the empirical scatter well. The most likely reconciliation: v (found by minimizing over the full 17-dimensional $(\phi, \theta_{z0}, \theta_{z100})$ space) is a mixed direction, not aligned with any single physical θ component’s own marginal-variance direction, so a 2–4 $\times$ local mis-estimate of curvature along v does not necessarily translate into a comparable error in any individual component’s marginal variance after the Schur-complement/inversion mixes all 17 directions together. This is a hypothesis, not something checked here – flagged in §6 as follow-up.

4 Step 2+3: Full-covariance CRB vs. empirical RMSE

For the same 10 shots, computed `full_per_shot_covariance(H)` and extracted $\sigma_{\theta_{z0}}, \sigma_{\theta_{z100}}$ (marginal std per component), aggregated as root-mean-square across shots. Compared against the empirical RMSE of $\hat{\theta}_{\text{best}} - \theta_{\text{true}}$, pooled over $z0$ and $z100$, from the 13 completed runs of the real `confocal_random_zernike` job available at the time of this check (5200 shot/z-slice points).

One data-quality issue found and handled: the raw empirical RMSE for σ_{vy} was 2.33×10^{-6} , a 3.46 $\times$ mismatch against the CRB’s 6.75×10^{-7} – while every other component matched to a few percent. Investigated directly rather than accepted: of 5200 points, a single shot has a σ_{vy} residual of 1.6×10^{-4} (161 μm) – roughly 70 $\times$ larger than the next-largest residual in that component and wildly inconsistent with the rest of the distribution (median $|\text{residual}| = 4.4 \times 10^{-7}$, matching σ_{vx} ’s median almost exactly). RMSE is quadratic in residuals, so this one likely optimizer failure (an L-BFGS-B run landing in a bad basin for that one shot – not a CRB or model problem) was enough to dominate the whole-sample RMSE by itself. Excluding it (a MAD-based robust-outlier filter, $> 20\sigma_{\text{robust}}$ in any component; 1/5200 points excluded) brings σ_{vy} ’s empirical RMSE to 6.77×10^{-7} .

Component	CRB (avg $z0, z100$)	Empirical RMSE (outlier excl.)	Ratio
μ_{x0}	9.655×10^{-6}	9.816×10^{-6}	1.017
μ_{y0}	9.655×10^{-6}	9.531×10^{-6}	0.987
μ_{vx0}	2.471×10^{-6}	2.511×10^{-6}	1.016
μ_{vy0}	2.473×10^{-6}	2.437×10^{-6}	0.985
σ_x	9.500×10^{-6}	9.912×10^{-6}	1.043
σ_y	9.461×10^{-6}	9.892×10^{-6}	1.046
σ_{vx}	6.767×10^{-7}	6.861×10^{-7}	1.014
σ_{vy}	6.745×10^{-7}	6.771×10^{-7}	1.004

Table 1: Full-covariance theta CRB vs. empirical RMSE (outlier-excluded), all 8 kinematic components, `confocal_random_zernike` dataset. All values in metres (or the matching velocity unit). Every component agrees to within $\pm 5\%$.

Result: the full-covariance CRB matches the empirical scatter to within a few percent, across all 8 components, once one known bad fit is excluded. This directly refutes the presumption (reasonable, given the prior 66–220× failure) that Fisher/CRB simply cannot predict θ ’s recoverability – it was never actually tried with the full covariance before now.

5 Framing: this is a single-shot comparison, and that’s intentional

θ is a fresh, independent nuisance parameter drawn from the prior *every shot* – it is not a shared, estimated-once parameter like $\beta = (A_s, A_c)$. Consequently, pooling more shots does *not* reduce the true per-shot θ RMSE (there is nothing to pool: shot i ’s θ_i tells you nothing about shot j ’s θ_j). What more shots *do* improve is the *measurement precision of the empirical RMSE estimate itself* – with 5200 points the empirical RMSE’s own relative uncertainty is $\sim 1/\sqrt{2 \times 5200} \approx 1\%$, tight enough to distinguish the observed few-percent CRB agreement from noise. This is why this comparison uses many shots (for a trustworthy *measurement*) while still being a fundamentally single-shot statement (the CRB itself is a per-shot quantity, not one that shrinks with N_{shots}) – unlike β ’s CRB, which genuinely does improve with shot count (`crb_signal.py`’s `crb_from_j/simple_scaling_crb`).

6 Open questions / future work

- **Reconcile §3 and §4.** Check what the worst-eigenvalue direction v actually projects onto in θ -component space, for a few shots – confirm (or refute) the “mixed direction, doesn’t dominate any single marginal” hypothesis directly rather than leaving it as speculation.
- **Gaussian-beam baseline comparison** (explicitly deferred from this note): repeat §4 for the matched unaberrated dataset already on disk, `data/R80_N200_A1000000_...f0.3000` (no `_psmap` suffix $\Rightarrow$ default `CONFOCAL.FINE`), which shares exactly the same n_{atoms} , cloud prior, and shot count as `confocal_random_zernike` – needs a modest (not full 20×200) `generate_kinematic_estimates.py` run to get empirical $\hat{\theta}_{\text{best}}$ scatter for it.
- **Connect to the β (signal) CRB.** If the aberrated case’s theta-CRB differs measurably from the Gaussian case’s (once the above is done), propagate that through `crb_signal.py`’s existing $j_{\text{eff}} \rightarrow$ beta-CRB machinery to get an actual quantitative bound on how much wavefront aberration/curvature costs in signal-extraction precision – the original motivating question for this whole line of work.
- The current check used only `run_000`’s first 10 shots for the CRB side (cheap: ~ 1 minute for all of §3–4 combined). Widening this to more shots/runs would tighten the CRB-side estimate too (currently it isn’t the limiting source of noise in the comparison, but worth checking as the empirical run finishes all 20 runs).